

Conditional Quantile Modeling of Financial Risk: Evidence from SPY Returns

Chris Hsueh

December 31, 2025

1 Introduction

Value-at-Risk (VaR) is a standard measure of market risk that summarizes the conditional lower tail of portfolio returns at a specified confidence level over a fixed horizon [Engle and Manganelli, 2004]. This report studies the problem of forecasting conditional lower quantiles of daily SPY returns using realized volatility measures, with the goal of assessing short-horizon downside risk. Because financial returns exhibit heavy tails, volatility clustering, and time-varying higher moments, credible VaR forecasting requires models that can adapt to changing risk conditions rather than relying on homoskedastic or Gaussian assumptions. In practice, inaccurate VaR forecasts can translate into distorted capital buffers and misleading assessments of downside exposure, particularly during periods of market stress.

Quantile regression offers a complementary route by estimating conditional quantiles directly, without requiring a fully specified return distribution. By modeling $Q_{r_{t+1}}(\alpha | \mathcal{F}_t)$ as a function of observable predictors, quantile regression targets the object of interest for risk management—the α -level tail—while remaining robust to heteroskedasticity and allowing covariates to have asymmetric effects across quantiles. These properties make it particularly suitable for short-horizon VaR forecasting at levels such as 1%, 5%, and 10%.

In this report, I study one-day-ahead VaR forecasting for SPY logarithmic returns using linear quantile regression with realized volatility measures as predictors. Forecasts are produced in a walk-forward out-of-sample setting, and performance is assessed using standard VaR backtests and proper scoring rules, including empirical violation rates, the Kupiec unconditional coverage test, the Christoffersen conditional coverage test, and the quantile (pinball) loss. For context, I also compare the quantile-regression-based VaR forecasts to those generated by a GARCH model and a neural quantile regression specification, highlighting differences in calibration, exceedance dynamics, and quantile forecast accuracy.

2 Dataset Description

The dataset consists of daily SPY log returns and two realized volatility measures derived from high-frequency intraday data. Realized volatility captures total intraday variation, while bipower variation isolates the continuous component, allowing separation of diffusive and jump-driven risk.

While daily returns exhibit little predictability in their conditional mean, their conditional variance displays strong persistence and asymmetry, with negative returns exerting a larger impact on future volatility than positive returns of equal magnitude.

The empirical analysis is conducted using daily data for the SPDR S&P 500 ETF (SPY). The dataset contains three key variables:

- Logarithmic returns r_t , representing daily asset returns.
- Realized volatility $rv5_t$, serving as a nonparametric proxy for daily integrated variance.
- Bipower variation bv_t , capturing the continuous component of price variation and providing robustness to jumps.

All observations with missing values are removed prior to model estimation to ensure consistency across predictors and targets.

2.1 Data Visualization and Characteristics

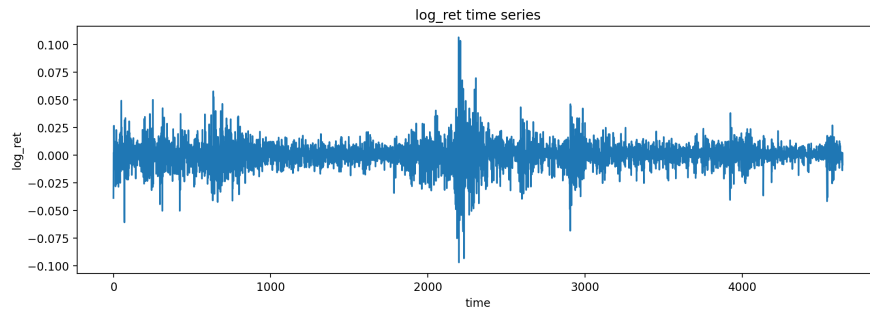


Figure 1: Log daily returns (r_t)

Preliminary exploratory analysis reveals several stylized facts that are well documented in the financial econometrics literature and are clearly present in the SPY dataset.

First, the time series of daily log returns exhibits pronounced volatility clustering (Figure 1), with extended periods of low volatility interspersed with episodes of extreme market stress. These high-volatility regimes coincide with sharp spikes in realized volatility measures, indicating that large absolute returns tend to cluster in time rather than occur independently.

Second, the unconditional distribution of returns is strongly non-Gaussian. As shown in Figure 2, quantile–quantile comparisons against the normal distribution suggest substantial deviations in both tails, with particularly heavy left-tail behavior. This is consistent with the high empirical kurtosis of returns and indicates that extreme losses occur more frequently than would be implied by a normal distribution, which favors the Student-t for the most realistic fit to downside risk.

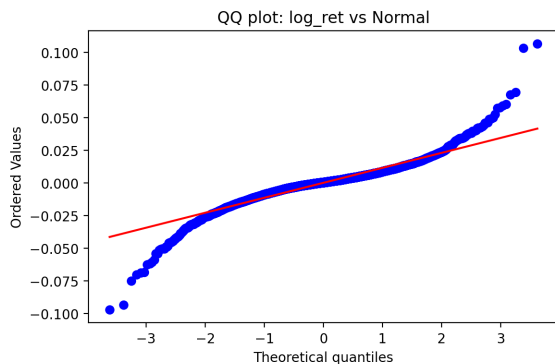


Figure 2: Quantile–quantile comparisons against the normal distribution (redline)

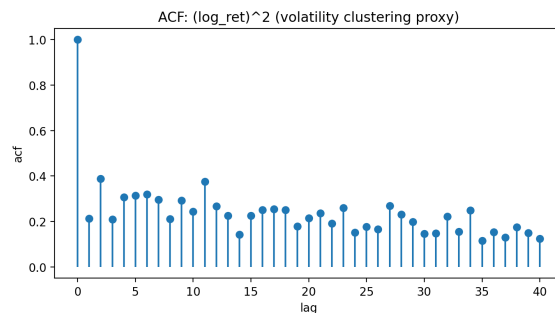


Figure 3: Autocorrelation function of squared log returns

Third, the squared return series exhibits persistent autocorrelation over many lags, providing clear evidence of conditional heteroskedasticity (Figure 3). This persistence supports the use of GARCH-type models that explicitly allow the conditional variance to evolve over time, rather than assuming homoskedastic or independently distributed returns.

The realized measures indicate that the dataset is of sufficient quality for subsequent econometric analysis and no significant pre-processing step is needed.

3 Methodology

3.1 Forecasting Framework

Let r_t denote the logarithmic return of SPY on day t . The objective is to forecast the conditional lower quantiles of the next-day return r_{t+1} given the information available at time t . Formally, for a

confidence level $\alpha \in (0, 1)$, the α -level Value-at-Risk is defined as the conditional quantile:

$$\text{VaR}_{t+1}(\alpha) = Q_{r_{t+1}}(\alpha \mid \mathcal{F}_t)$$

where $\mathcal{F}_t$ denotes the information set at time t .

The forecast target is defined as:

$$y_t = r_{t+1}$$

, and the vector of predictors observable at time t is given by:

$$X_t = [1 \ r_t \ \text{rv}5_t \ \text{bv}_t]$$

where an intercept is included, r_t is the contemporaneous return, $\text{rv}5_t$ is realized volatility, and bv_t is bipower variation. This specification ensures that the forecasting exercise is free from look-ahead bias.

3.2 Linear Quantile Regression

Quantile regression provides a direct approach to modeling conditional quantiles of the response variable without imposing parametric assumptions on the full conditional distribution. For a given quantile level α , the linear quantile regression model is specified as:

$$Q_{y_t}(\alpha \mid X_t) = X_t^\top \beta(\alpha)$$

where $\beta(\alpha)$ is a vector of quantile-specific coefficients.

The parameter vector $\beta(\alpha)$ is estimated by solving:

$$\hat{\beta}(\alpha) = \arg \min_{\beta} \sum_t \rho_{\alpha}(y_t - X_t^\top \beta)$$

where $\rho_{\alpha}(u)$ denotes the check (pinball) loss function:

$$\rho_{\alpha}(u) = u(\alpha - \mathbb{I}\{u < 0\})$$

This loss function asymmetrically penalizes underprediction and overprediction, making it well suited for estimating lower-tail risk measures such as VaR.

3.3 GARCH-Based Value-at-Risk

As a benchmark from the parametric volatility modeling literature, Value-at-Risk (VaR) forecasts are also generated using a GARCH framework. GARCH-type models are designed to capture conditional heteroskedasticity in financial returns by allowing the conditional variance to evolve dynamically as a function of past shocks and past volatility. This feature makes them a natural reference point for VaR forecasting.

Let z_t be an i.i.d. innovation with zero mean and unit variance. The conditional mean–variance representation is then given by:

$$r_t = \mu_t + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t.$$

The conditional variance σ_t^2 follows a GARCH(p, q) process:

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2,$$

with parameters satisfying standard positivity and stationarity conditions.

Given estimates of the conditional mean and variance, the one-step-ahead VaR at level α is computed as

$$\text{VaR}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} F_z^{-1}(\alpha),$$

where $F_z^{-1}(\alpha)$ denotes the α -quantile of the assumed innovation distribution. To accommodate heavy tails commonly observed in equity returns, Student- t innovations are employed. Model parameters are re-estimated in a walk-forward manner to ensure consistency with the real-time forecasting setting.

3.4 Neural Quantile Regression

To assess whether nonlinear function approximation improves tail risk forecasts, a neural quantile regression model is considered as an additional comparison method. Unlike linear quantile regression, this approach allows for flexible nonlinear interactions between predictors and the conditional quantiles of future returns.

The neural quantile regression model is implemented as a multilayer perceptron with two hidden layers and ReLU activations. Let $X_t \in \mathbb{R}^3$ denote the predictor vector $(r_t, rv5_t, bv_t)$. The network produces K raw outputs $(z_1, \dots, z_K)$, which are transformed into ordered conditional quantiles via a cumulative construction:

$$\hat{q}_1 = z_1, \quad \hat{q}_k = z_1 + \sum_{j=2}^k \text{softplus}(z_j), \quad k = 2, \dots, K.$$

This guarantees non-crossing quantiles for all inputs by construction.

Model estimation follows a rolling-window, one-step-ahead forecasting scheme. For each forecast origin t , the model is trained using observations contained in the current estimation window, and the conditional quantiles for r_{t+1} are obtained by a forward pass through the network. Input features are standardized using z-score normalization, with the mean and standard deviation computed only within the training window and then applied to the corresponding forecast observation, preventing look-ahead bias.

The training objective is the sum of pinball losses across all quantile levels,

$$\mathcal{L}_t = \sum_{k=1}^K \rho_{\alpha_k}(y_t - \hat{q}_k(X_{t-1})),$$

which is optimized using the Adam algorithm with ℓ_2 weight regularization. Mini-batch stochastic gradient descent is employed, and early stopping based on training loss is applied during the initial estimation phase to stabilize convergence.

To reduce computational cost while maintaining the walk-forward structure, the network parameters are warm-started between consecutive forecast steps. To mitigate potential accumulation of local optimization bias, the network is periodically reinitialized and retrained from scratch at fixed intervals.

3.5 Experiment Settings

3.5.1 Model Specification

VaR forecasts are generated using linear quantile regression, estimated separately for each quantile level $\alpha \in \{0.01, 0.05, 0.10\}$. For a given quantile level α , the conditional quantile function of next-day returns is specified as

$$Q_{r_{t+1}}(\alpha | X_t) = X_t^\top \beta_\alpha,$$

where X_t denotes the vector of predictors observed at time t and β_α is a quantile-specific coefficient vector.

Model estimation is conducted using the linear quantile regression framework implemented in `statsmodels`. A sufficiently large iteration cap (`max_iter=1000`) is employed to ensure numerical convergence of the optimization routine.

For the comparison methods: GARCH (1,1) model is used, and the neural quantile regression model has 2 hidden layers.

3.5.2 Evaluation and Backtesting

Forecast performance is assessed using standard VaR backtesting procedures applied to the out-of-sample forecasts generated by the walk-forward scheme. Specifically, the following evaluation metrics and statistical tests are employed:

- **Violation indicators**, unconditional calibration via the empirical violation rate, defined as realizations where the observed return falls below the predicted VaR.

- **Kupiec unconditional coverage test**, assessing whether the observed violation rate is consistent with the theoretical level α .
- **Christoffersen conditional coverage test**, jointly testing correct unconditional coverage and the independence of violations over time [Christoffersen, 1998].
- **Quantile (pinball) loss**, used as a proper scoring rule to evaluate the accuracy of quantile forecasts.

4 Results

Table 1 summarizes out-of-sample VaR backtesting results for SPY returns at confidence levels $\alpha \in \{0.01, 0.05, 0.10\}$ under three forecasting approaches. In what follows, a model is considered statistically adequate at a given α when the null hypotheses of correct unconditional and conditional coverage are not rejected at conventional significance levels ($p < \alpha$). (It’s worth saying that in VaR backtesting, large p-values indicate that the null hypothesis cannot be rejected. However, p-values are not measures of model accuracy, and excessively conservative VaR forecasts may also yield large p-values.)

Table 1: VaR Backtesting Results for SPY Returns

Model	α	Violation Rate	Kupiec p	Christoffersen p	Quantile Loss
Quantile Regression	0.01	0.0143	0.0219	0.1681	0.000396
	0.05	0.0503	0.9315	0.7163	0.001287
	0.10	0.0943	0.2827	0.5485	0.002055
GARCH	0.01	0.0099	0.9441	0.0379	0.000361
	0.05	0.0446	0.1577	0.4798	0.001250
	0.10	0.0803	0.0001	0.8544	0.002003
Neural Quantile Regression	0.01	0.0073	0.1141	0.1616	0.001038
	0.05	0.0593	0.0206	0.00003	0.001668
	0.10	0.1156	0.0043	0.2305	0.002401

4.1 Backtesting Interpretation

Let $\widehat{\text{VaR}}_{t+1}(\alpha)$ denote the one-step-ahead VaR forecast and let r_{t+1} denote the realized next-day return. A VaR violation (exceedance) occurs when $r_{t+1} < \widehat{\text{VaR}}_{t+1}(\alpha)$. The violation indicator is

$$I_{t+1}(\alpha) = \mathbb{I}\{r_{t+1} < \widehat{\text{VaR}}_{t+1}(\alpha)\}.$$

Under correct calibration, $\mathbb{E}[I_{t+1}(\alpha)] = \alpha$ and the sequence $I_{t+1}(\alpha)$ should not exhibit systematic temporal dependence.

4.2 Linear Quantile Regression

Linear quantile regression performs strongly at moderate tail probabilities. At $\alpha = 0.05$, the empirical violation rate is 0.0503, closely matching the nominal level, and both tests fail to reject correct coverage (Kupiec $p = 0.9315$; Christoffersen $p = 0.7163$). At $\alpha = 0.10$, the violation rate is 0.0943 and the null hypotheses of correct unconditional and conditional coverage are again not rejected (Kupiec $p = 0.2827$; Christoffersen $p = 0.5485$). These results indicate that the linear conditional-quantile specification, combined with realized volatility measures, provides a coherent and stable characterization of downside risk for the 5% and 10% VaR levels.

At the most extreme quantile $\alpha = 0.01$, the violation rate rises to 0.0143, exceeding the nominal level and implying that the 1% VaR is not sufficiently conservative on average. This miscalibration is reflected in rejection of unconditional coverage (Kupiec $p = 0.0219$). In contrast, conditional coverage is not rejected (Christoffersen $p = 0.1681$), suggesting that the main deficiency at $\alpha = 0.01$ is the level of the tail forecast rather than pronounced violation clustering.

4.3 GARCH

The GARCH-based model achieves excellent unconditional calibration at $\alpha = 0.01$. The violation rate is 0.0099, and the Kupiec test provides no evidence against correct unconditional coverage (Kupiec $p = 0.9441$). However, conditional coverage is rejected at conventional levels (Christoffersen $p = 0.0379$), indicating temporal dependence in exceedances. Empirically, this pattern is consistent with a model that matches the long-run frequency of tail events while failing to fully account for the dynamics governing the timing of those events.

At $\alpha = 0.05$, the model is mildly conservative (violation rate 0.0446) and neither test rejects (Kupiec $p = 0.1577$; Christoffersen $p = 0.4798$). At $\alpha = 0.10$, the model becomes substantially conservative: the violation rate drops to 0.0803 and unconditional coverage is strongly rejected (Kupiec $p = 0.0001$), while conditional coverage is not rejected (Christoffersen $p = 0.8544$). Thus, at the 10% level the GARCH specification tends to overstate risk by producing VaR forecasts that are too negative.

In terms of the proper scoring rule, GARCH attains the lowest pinball loss across all α in Table 1. The gains over linear quantile regression are modest but consistent, indicating slightly improved average quantile accuracy despite the calibration shortcomings at $\alpha = 0.10$.

4.4 Neural Quantile Regression

Neural quantile regression does not deliver robust improvements in this experiment. At $\alpha = 0.01$, the violation rate is 0.0073, implying a conservative lower-tail forecast, with no rejection by either test (Kupiec $p = 0.1141$; Christoffersen $p = 0.1616$). Nevertheless, the pinball loss is substantially larger (0.001038) than for the other models, indicating inferior quantile accuracy and reduced sharpness.

At $\alpha = 0.05$, performance deteriorates materially: the violation rate increases to 0.0593, and both tests reject (Kupiec $p = 0.0206$; Christoffersen $p = 0.00003$). The extremely small Christoffersen p -value provides strong evidence of violation dependence, consistent with clustered exceedances. At $\alpha = 0.10$, the violation rate rises to 0.1156 and unconditional coverage is again rejected (Kupiec $p = 0.0043$). Conditional coverage is not rejected at $\alpha = 0.10$ (Christoffersen $p = 0.2305$), but the systematic over-violation indicates that the predicted 10% quantile is too high on average.

4.5 Overall Comparison

Taken together, the evidence supports linear quantile regression as a stable and interpretable main method for moderate tail risk. It passes both backtests at $\alpha = 0.05$ and $\alpha = 0.10$ and achieves competitive pinball loss. Its principal limitation is at $\alpha = 0.01$, where unconditional coverage is rejected.

GARCH provides the most accurate scoring-rule performance and near-perfect unconditional calibration at $\alpha = 0.01$, but it fails conditional coverage at the 1% level and becomes overly conservative at $\alpha = 0.10$. Neural quantile regression exhibits systematic miscalibration and violation clustering at $\alpha = 0.05$ and does not outperform the simpler alternatives. Overall, the results suggest that realized volatility information is valuable for tail forecasting, but increased model flexibility does not automatically translate into superior VaR performance in this setting.

5 Discussion and Conclusion

5.1 Discussion

This report evaluates one-day-ahead VaR forecasts for SPY returns using realized volatility measures as predictors, with linear quantile regression as the main method and GARCH and neural quantile regression as comparison benchmarks. The empirical results highlight two recurring themes in short-horizon tail forecasting: (i) moderate-tail VaR levels (e.g., 5% and 10%) can often be modeled reliably with relatively simple specifications, while (ii) the extreme tail (e.g., 1%) is substantially harder to calibrate due to scarcity of tail observations and stronger nonlinearity during stress periods.

Linear quantile regression delivers statistically coherent performance at moderate tail probabilities, and it provides a transparent mapping from realized measures to downside risk, which is attractive for interpretability and for diagnosing which predictors are informative for tail outcomes. The primary weakness arises at the most extreme quantile level, where unconditional coverage can be rejected even

when exceedances do not exhibit strong evidence of clustering. Empirically, this pattern is consistent with a model that captures time variation in risk but remains insufficiently conservative in the far left tail.

The GARCH benchmark illustrates the distinction between matching the *frequency* of tail events and capturing their *timing*. The results reflect a practical limitation of volatility-driven parametric VaR: mapping conditional variance to a fixed innovation quantile does not guarantee that the exceedance process is dynamically well behaved, particularly when tail risk depends on additional state variables (e.g., leverage effects, jumps, or regime changes).

Neural quantile regression does not deliver robust improvements in this experiment. Despite its flexibility, the neural model exhibits instability and miscalibration at moderate quantiles, together with evidence consistent with clustered exceedances. A plausible interpretation is that, given the limited predictor set and the effective sample size within each rolling window, additional functional flexibility increases estimation variance and susceptibility to overfitting, degrading out-of-sample calibration. This result emphasizes that model complexity is not inherently beneficial for VaR forecasting unless supported by richer features, stronger regularization, and careful validation under time-series constraints.

Overall, the evidence supports linear quantile regression as a credible and defensible main specification for moderate tail risk, with GARCH and neural alternatives serving as informative diagnostic benchmarks. The results also suggest that realized volatility information materially improves tail forecasts, but additional nonlinearity alone does not guarantee better backtesting performance.

5.2 Limitations

Several limitations should be noted. First, the predictor set is intentionally parsimonious, relying primarily on realized volatility measures and a lagged return; omitted state variables may contain incremental information for tail risk. Second backtesting focuses on unconditional and conditional coverage tests and pinball loss; these capture calibration and scoring-rule accuracy but do not fully characterize economic costs, such as capital efficiency or tail loss severity conditional on exceedance. Finally, rolling-window estimation introduces a bias–variance trade-off that depends on window length and structural stability; alternative window choices could affect comparative performance.

5.3 Conclusions and Future Work

The main conclusion is that a linear quantile regression framework, augmented with realized volatility measures, provides a stable and interpretable approach to one-day-ahead VaR forecasting for SPY at moderate tail probabilities. However, accurate calibration at the 1% level remains challenging, indicating that extreme downside risk may require either richer conditioning information or models specifically tailored to tail dynamics.

Several extensions are natural. First, the linear quantile regression can be enriched using additional realized measures (e.g., realized semivariance) and jump-robust components, or by allowing nonlinearities through splines while maintaining interpretability. Second, tail-specific approaches such as CAViaR-type dynamics or hybrids that combine volatility models with extreme value methods could improve calibration at very low quantiles [Engle and Manganelli, 2004]. Third, neural quantile models may benefit from stronger regularization, stricter time-series cross-validation, and feature expansion; without these, increased flexibility risks deteriorating out-of-sample reliability. These directions provide a coherent path to improving extreme-tail performance while preserving the methodological discipline required for risk management applications.

References

- [Christoffersen, 1998] Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4):841–862.
- [Engle and Manganelli, 2004] Engle, R. F. and Manganelli, S. (2004). Caviar: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4):367–381.